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Modeling of Serine Protease Substrate Catalysis In Synthetic Micelle Environment

Background

Catalysis occurring in micellar environments serves to accelerate biochemical reactions. Serine proteases, which are a large family of enzymes in which serine is used to attack the peptide bond in proteins, are essential in many biological processes across different organisms. For example, serine proteases like trypsin and chymotrypsin are vital for digestion, granzymes are essential in immune responses, and thrombin is important in blood clotting. Serine proteases are also implicated in many disease processes, so understanding the mechanisms of serine proteases and associated inhibitors could be useful in new therapeutic strategies for treating a wide variety of diseases.

Goals

This project aims to develop a machine learning-based simulation framework capable of using enzyme structures, ligand docking, and substrate information to predict catalytic efficiency within micelles. This machine learning framework will be able to model enzyme-substrate interactions within micelles, enzyme catalysis within micellar environments, binding affinity, and reaction dynamics.

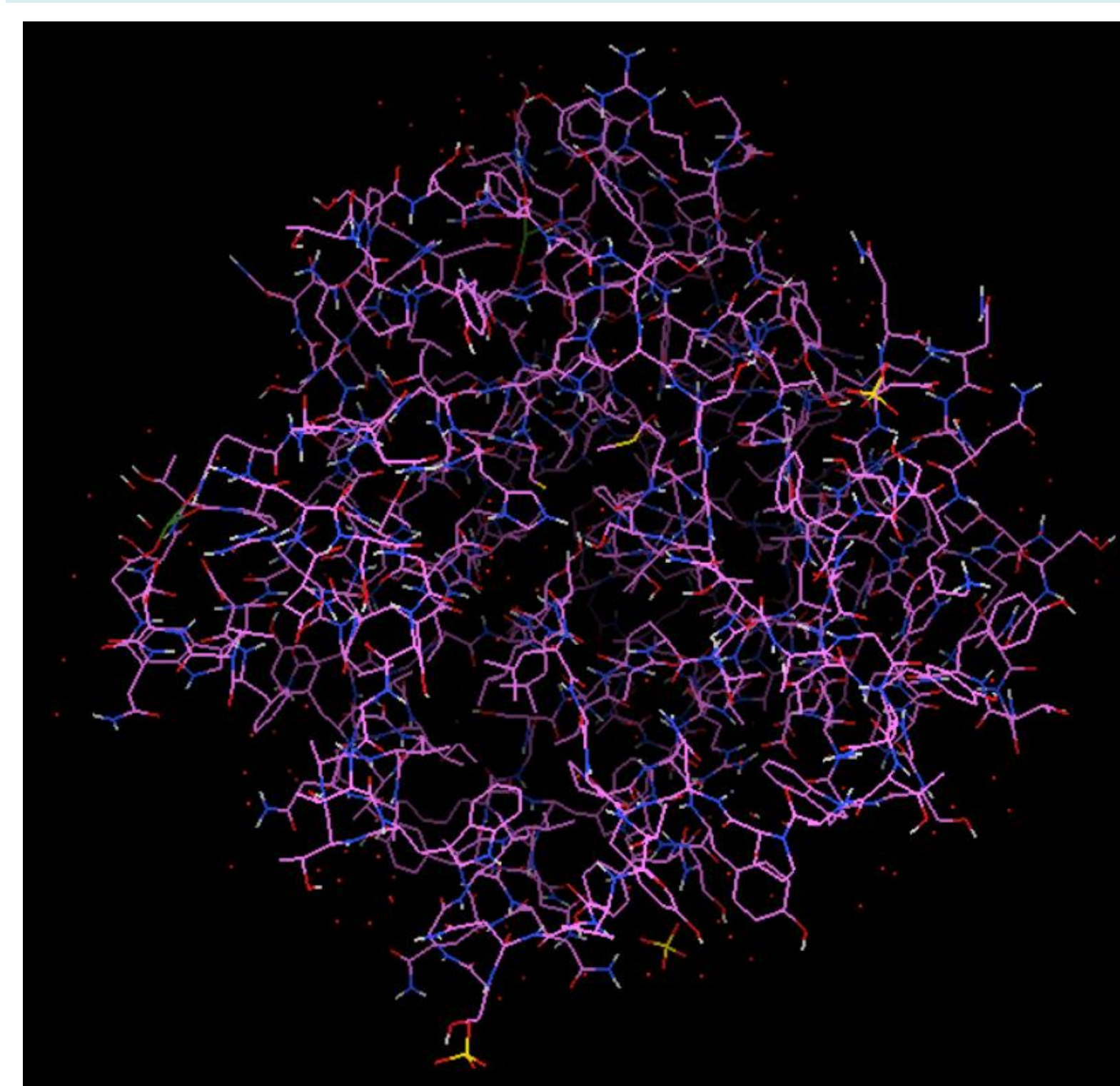


Figure 3. This displays the protein 1THM also known as thermitase. This enzyme is a serine protease used in the docking procedure highlighted in this poster.

Challenges

It was challenging to work with such a large set of data. Such a large dataset presented many possibilities for research process approaches. Additionally, converting so many molecules to sdf and PDBQT files took a lot of processing power and time.

Methods

A dataset of serine proteases structures from the Protein Data Bank was collected, along with a dataset of over 10,000 substrate molecules. Properties such as molecular weight, TPSA, and FSP3 were gathered for each molecule. Graphs were made in VS Code using Matplotlib and Plotly comparing many of the molecule's properties to identify trends. For docking purposes, each substrate molecule was converted via its SMILES notation into an sdf file, then into a PDBQT file. Batch docking was performed to obtain many results at once due to large amounts of data. Molecular docking identified binding poses within the enzyme's active site..

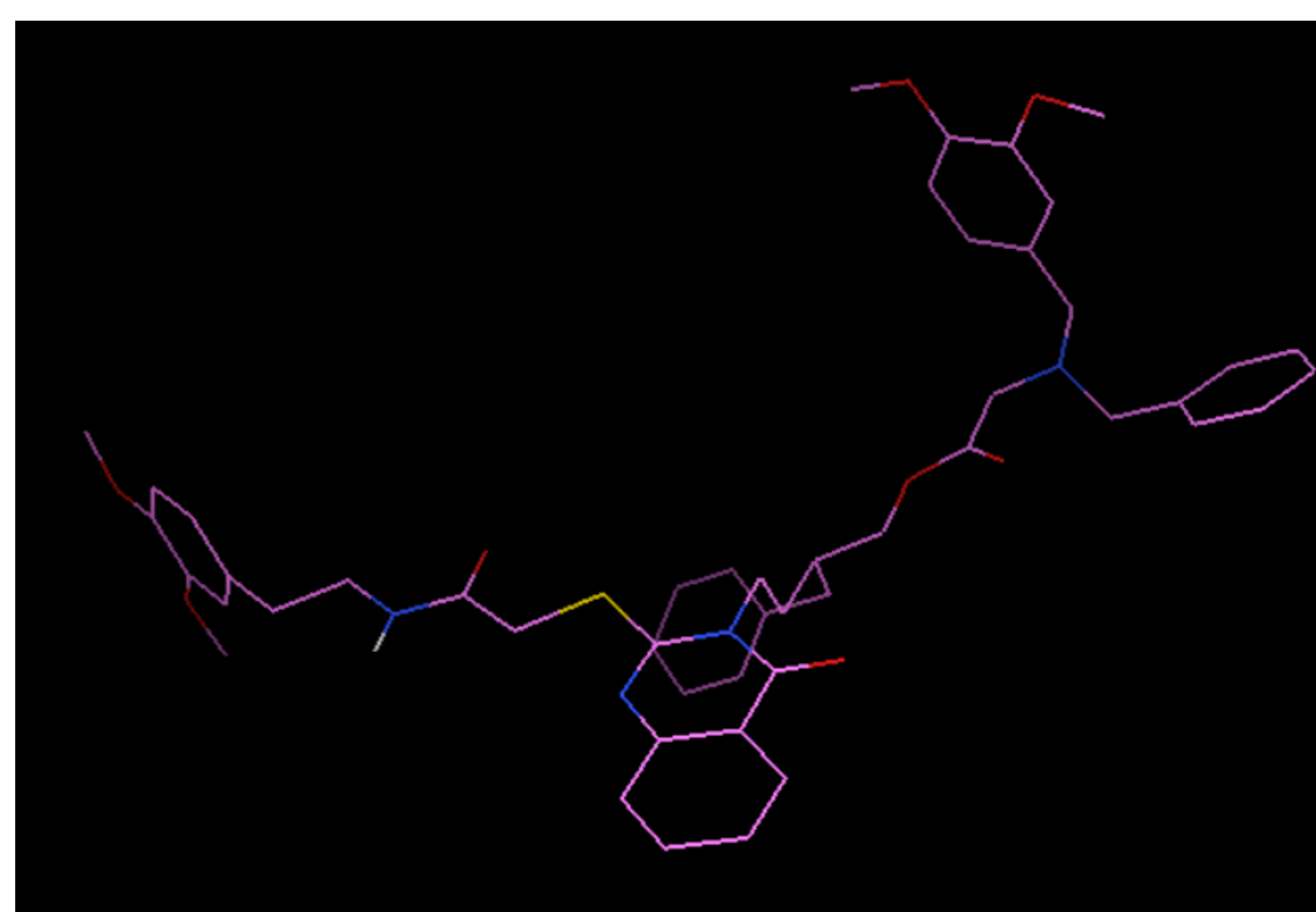


Figure 2. This figure displays the small molecule that acted as a ligand during one docking. At a molecular weight of 838.99 g/mol, N-[2-(3,4-dimethoxyphenyl)ethyl]-2-[[3-(3-{2-[2-(3,4-dimethoxyphenyl)ethylamino]-2-oxoethoxy}propyl)-1H-indol-5-yl]amino]-2-oxoethyl-N-(2-phenylethyl)acetamide was by far the heaviest small molecule, hence why it was chosen to be highlighted to see how it would bind to a protein.

Future Steps

There is a great deal of work still to be done on this project. Great progress has been made in terms of docking, but the next steps aim to:

- Train the model with labeled derived data from experimental kinetics and computational simulations
- Implement graph neural networks to learn substrate features and predict catalytic efficiency
- Perform molecular dynamic simulations of enzyme-ligand interactions within enzymes to study how environment influences binding affinity and transition state stabilization
- Compare computational predictions to available experimental data

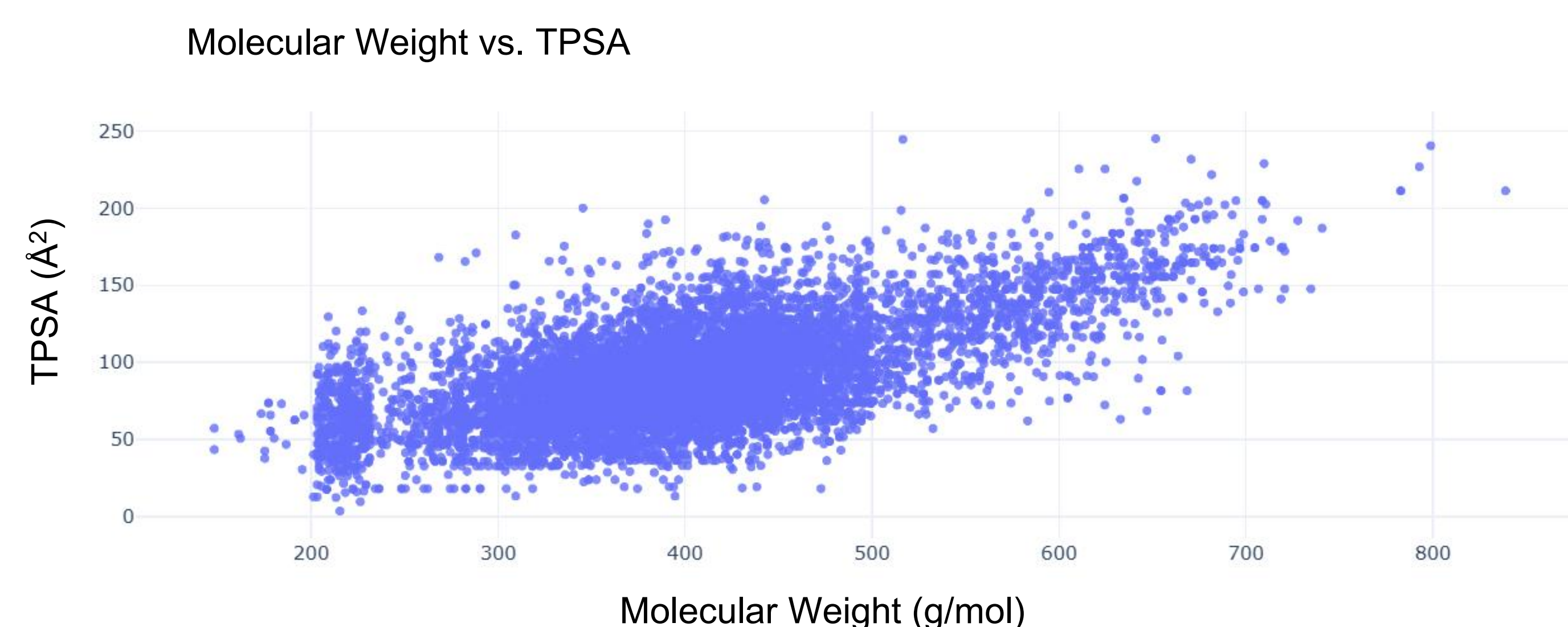


Figure 1. This figure shows an example graph generated from the small molecule database comparing all molecular weights in g/mol to all TPSA values in Å². A slight positive linear correlation can be seen between the two values. Over 50 graphs similar to Figure 1 were made comparing many different values of the small molecules in order to search for trends and outliers.

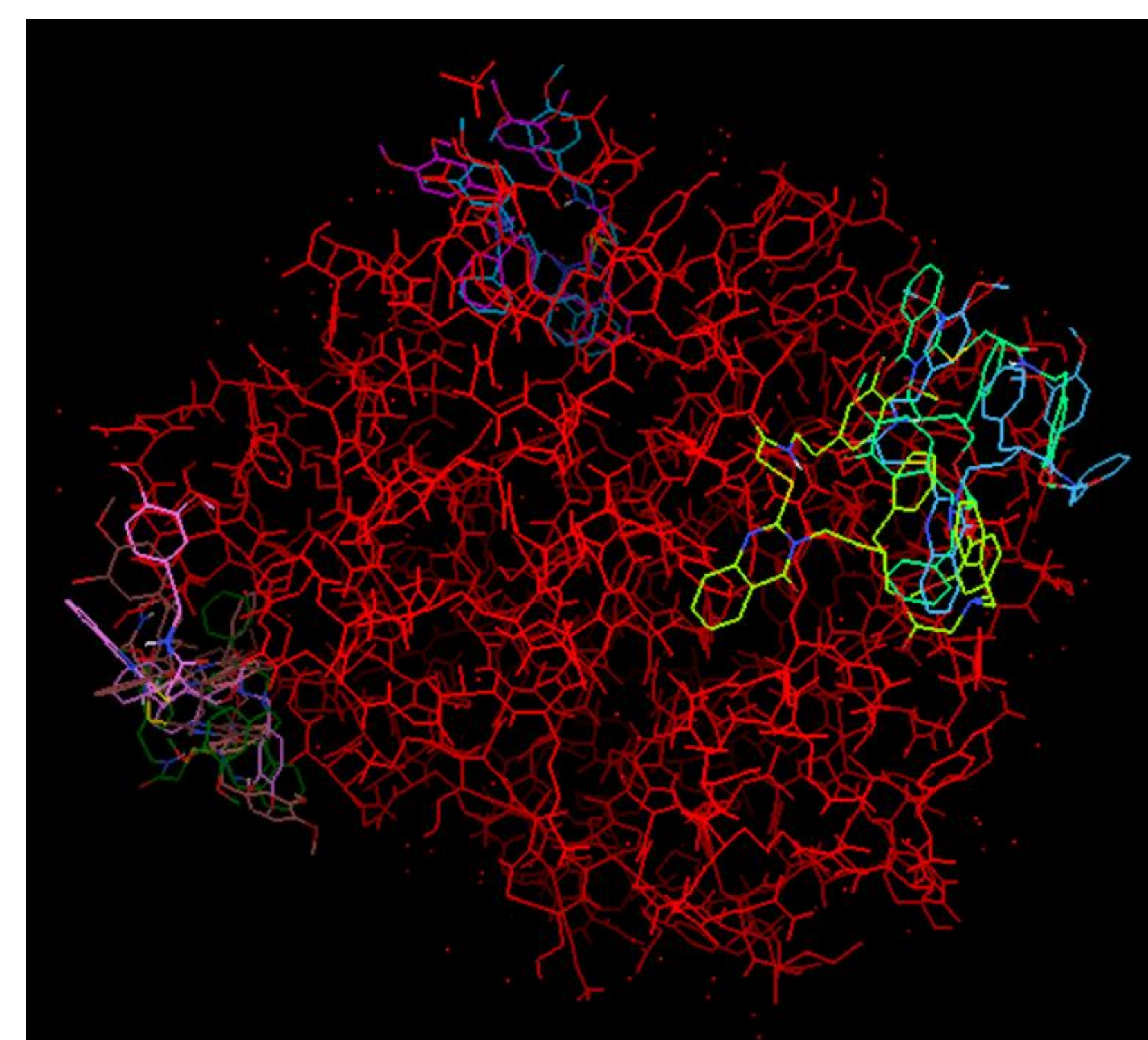


Figure 4. This image shows the final docking result of Figures 2 and 3. The protein is shown in red, and then each binding position of the ligand is in a different color in different areas of the protein. Each of these binding positions is a possible pose, with the binding affinity and RMSD of each pose being shown in Table 1 to determine which is most likely to be the best binding pose.

Pose	Binding Affinity (kcal/mol)	RMSD (lower)	RMSD (upper)
1	-5.450	0.000	0.000
2	-5.432	6.802	11.768
3	-5.226	7.688	11.956
4	-5.204	35.602	40.457
5	-5.193	2.878	12.817
6	-5.092	24.816	29.788
7	-5.060	24.907	29.805
8	-4.900	34.606	39.440
9	-4.875	30.609	35.772

Table 1. The data of table 1 displays results from the docking in Figure 4. A table like this can be generated for each docking. Pose 1 is the best binding position since it has the most negative affinity. RMSD, root mean square deviation, highlights the deviation from the best position. RMSD lower measures only the bare deviation from the best position, while RMSD accounts for flexibility of molecules.

Conclusion

Thousands of ligands were docked to serine proteases to generate a rich dataset of binding poses and affinities. Docking on a large scale enables deeper learning of structure-function relationships and is crucial in training ML models that can predict catalytic efficiency behavior in micellar environments.

References

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